



PF LAB 1

Student Lab Tasks — Problem Solving

Course Code: CL1002**Lab No: 01****Subject:** Programming Fundamental**Semester / Term:** 1st Semester**Instructor:** Mr. Shayan Ali

STUDENT SUBMISSION DETAILS

Name	Mokish sharma
Roll No / ID	26k-3147
Class / Section	BSE 1D
Date of Submission	27 AUGUST

Submission Guidelines

1. Submit your all tasks before date of submission
2. Insert Screenshot of your Scratch Problem (Code+Output)

TASK

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01 Student Result Percentage

Ali obtained **75, 82, and 68 marks** in three subjects. Each subject has a total of 100 marks. Find obtained marks and Percentage

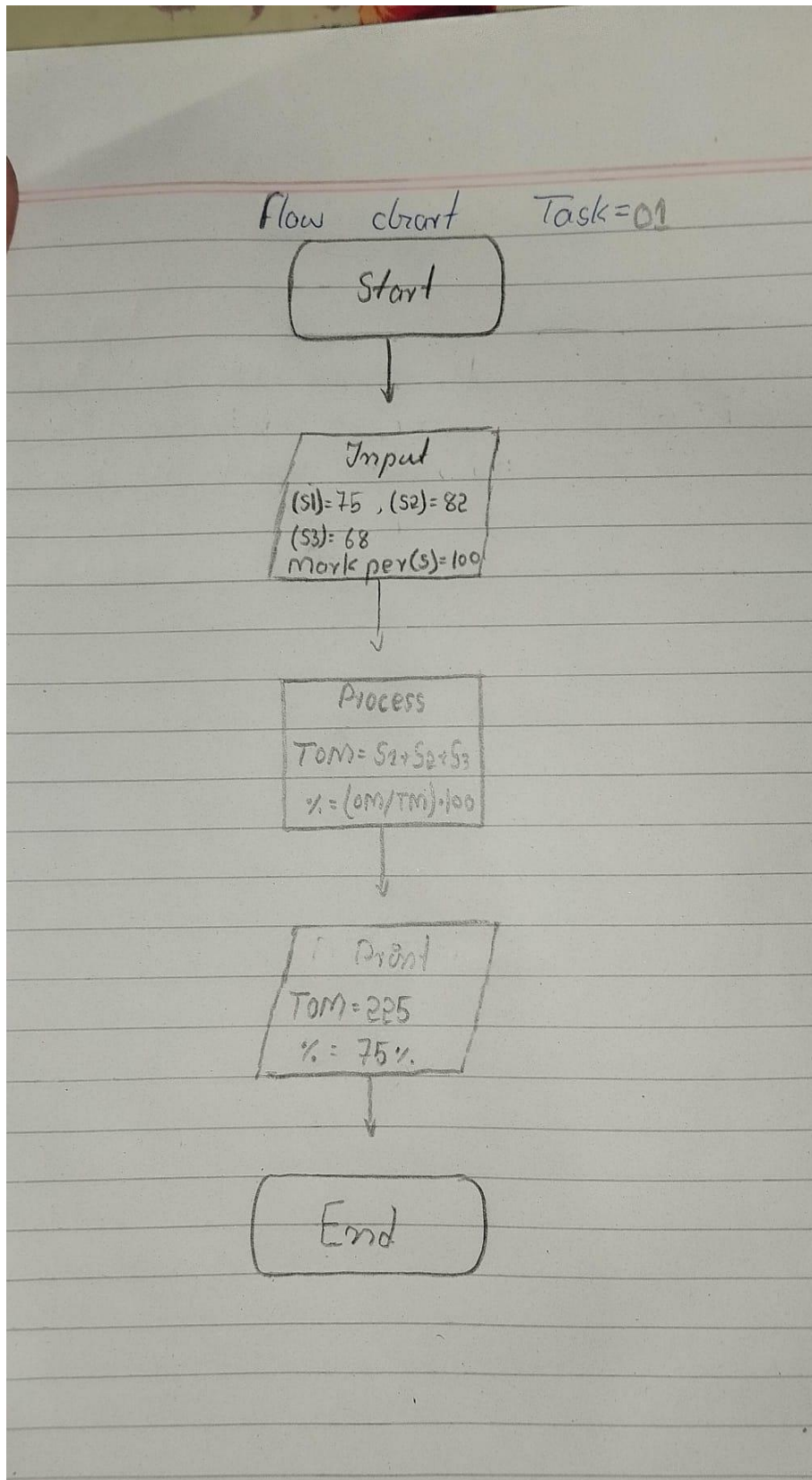
1. Create a PAC.

PAC Task = 01

Given data	Required-Result
Mark in (S1) = 75	Total Obtain
Mark in (S2) = 82	marks
Mark in (S3) = 68	Percentage
Marks per each subject = 100	
Processing - required	Solution - alternate
• $TOM = (S1) + (S2) + (S3)$	$TOM = 225$
	$\% = 75\%$
• $\% = (OM/TM) \cdot 100$	

TASK

2. Create Flow Chart



TASK

3. Create IPO

IPO Task = 02

Input	Processing	Module reference	Output
Mark in (S1) = 75	• $TOM = (S1) + (S3)$	Calcu	• $TOM = 225$
Mark in (S2) = 82	(S3)		• % = 75%
Mark in (S3) = 68	• $\% = (OM/TM) \cdot 100$	Calcu	
Mark per each (S) = 100			

02 Shopping Discount

Sara wants to buy a school bag costing **Rs. 2,000**. The shop is offering a **15% discount**. Find Discounted Amount

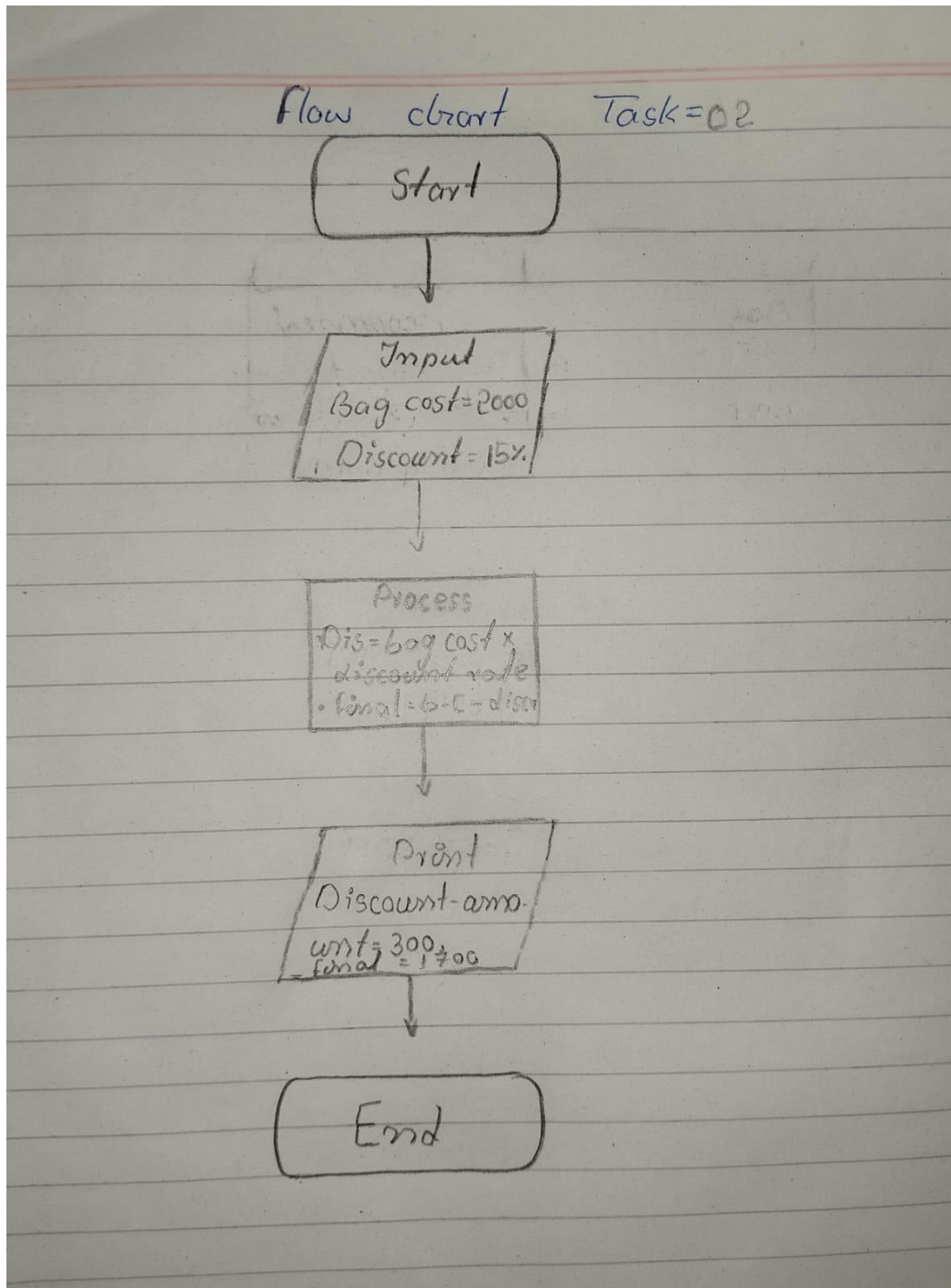
1. Create a PAC.

PAC Task = 02

Given data	Required-Result
Bag Cost = 2000	• Discounted Amount
Discount rate = 15%	• final amount
Processing - required	Solution - alternate
• $Dis = \text{Bg cost} \times \text{discount rate} / 100$	• Discounted amount = 300
• final amount = bag cost - discount	• final amount = 1700

TASK

2. Create Flow Chart



TASK

—

3. Create IPO

IPO Task = 02

Input	Processing	module reference	Output
• Bag Cost = 2000	• Dis = bag cost. x discount rate	• Calcu	• Dis amount = 300
• Dis rate = 15 %	• final am = bag cost - discount	• Calcu	• Final amount = 1700

TASK

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03 Water Tank Volume

water tank is **20 meters long, 10 meters wide, and 5 meters high**. Find volume

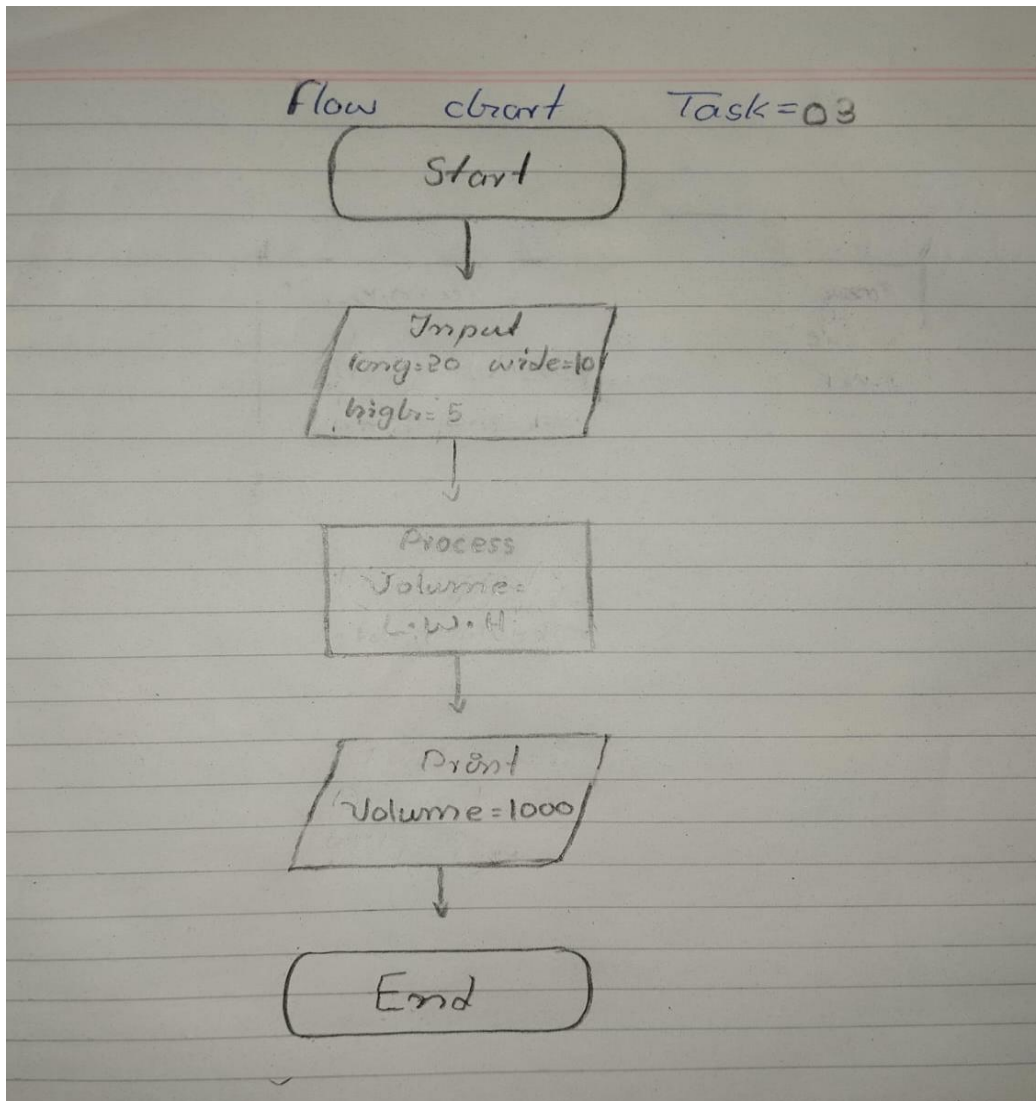
1. Create a PAC.

Given data	Required-Result
long = 20-m	• Volume
Wide = 10-m	
High = 5-m	
Processing-required	Solution-alternate
• Volume = L * W * H	• Volume =
	• 1000m³

TASK

A

2. Create Flow Chart



TASK

A

3. Create IPO

IPO Task = 03

Input	Processing	Module reference	Output
long = 20m	• Volume	• Calcu	Volume = 1000
Wide = 10m	$L \cdot W \cdot H$		
High = 5m			

TASK # —

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04 Rectangle Area

rectangular classroom board has a **length of 15 cm** and a **width of 8 cm**. Find Rectangle Area

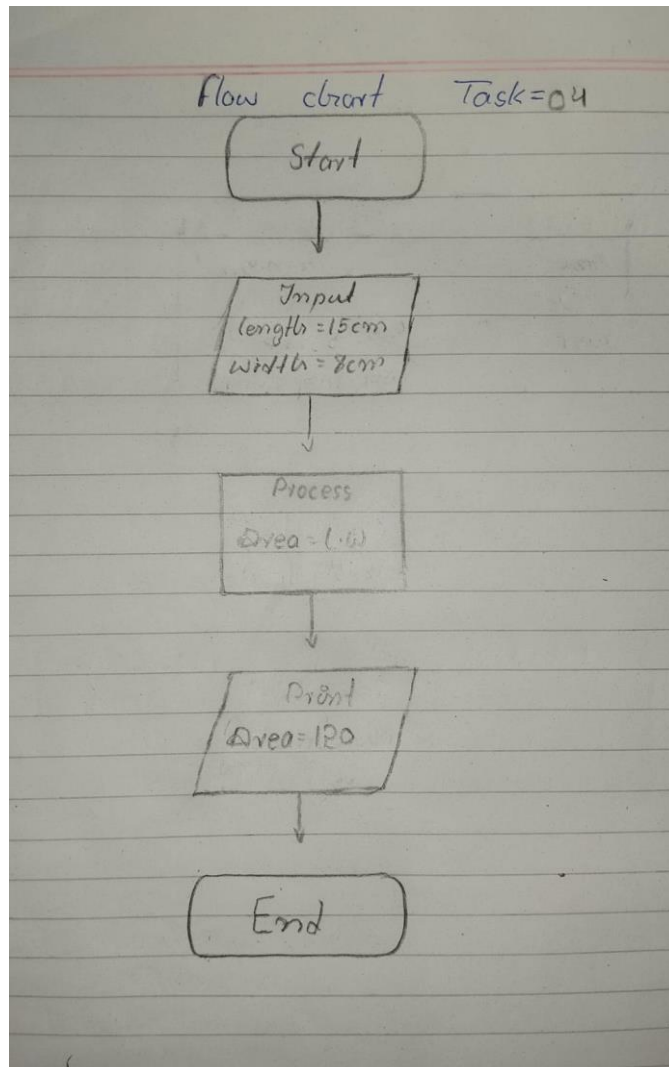
1. Create a PAC.

Given data	Required-Result
- length = 15 cm - width = 8 cm	Area
Processing - required	Solution - alternate
Area = 120	Area = 120

TASK # —

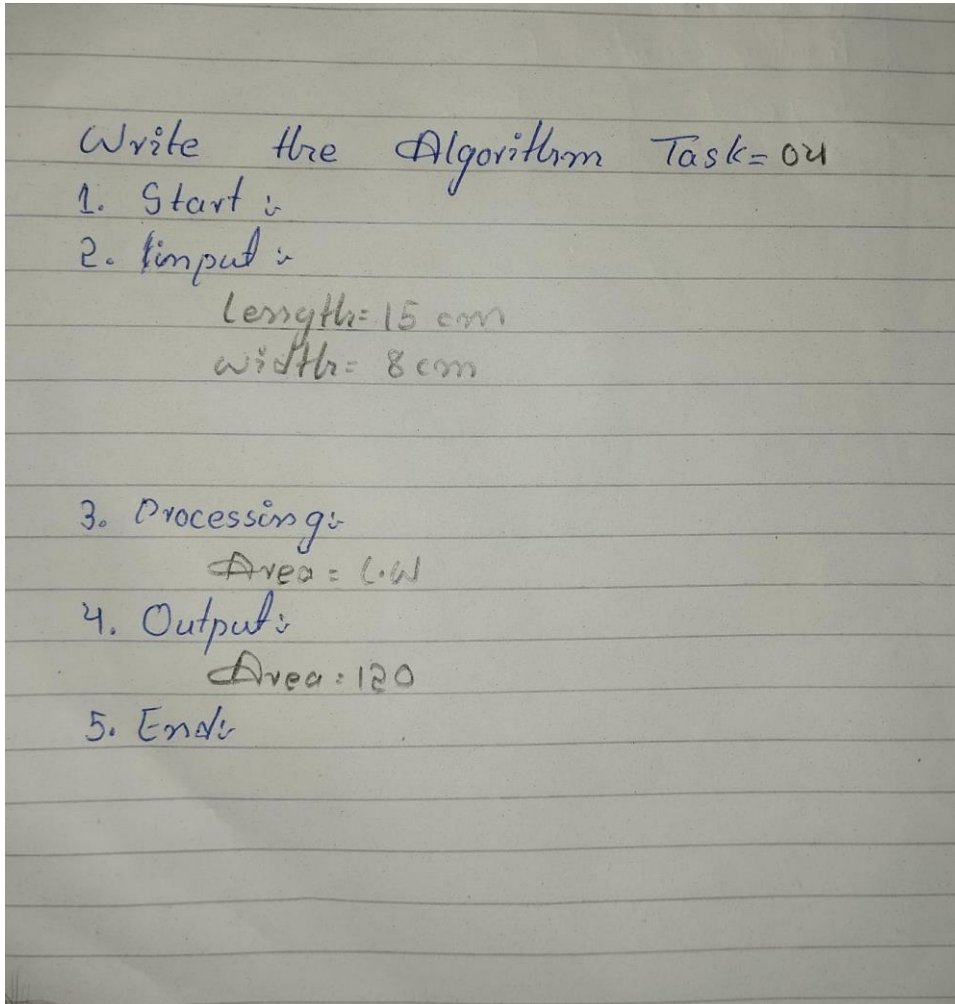
A

2. Create Flow Chart



TASK

A

3. Write an Algorithm.

TASK

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A

05 University Field Trip

university bus travels 12 km north, 8 km east, 10 km south, and 5 km west during a field trip. Find Total distance travelled.

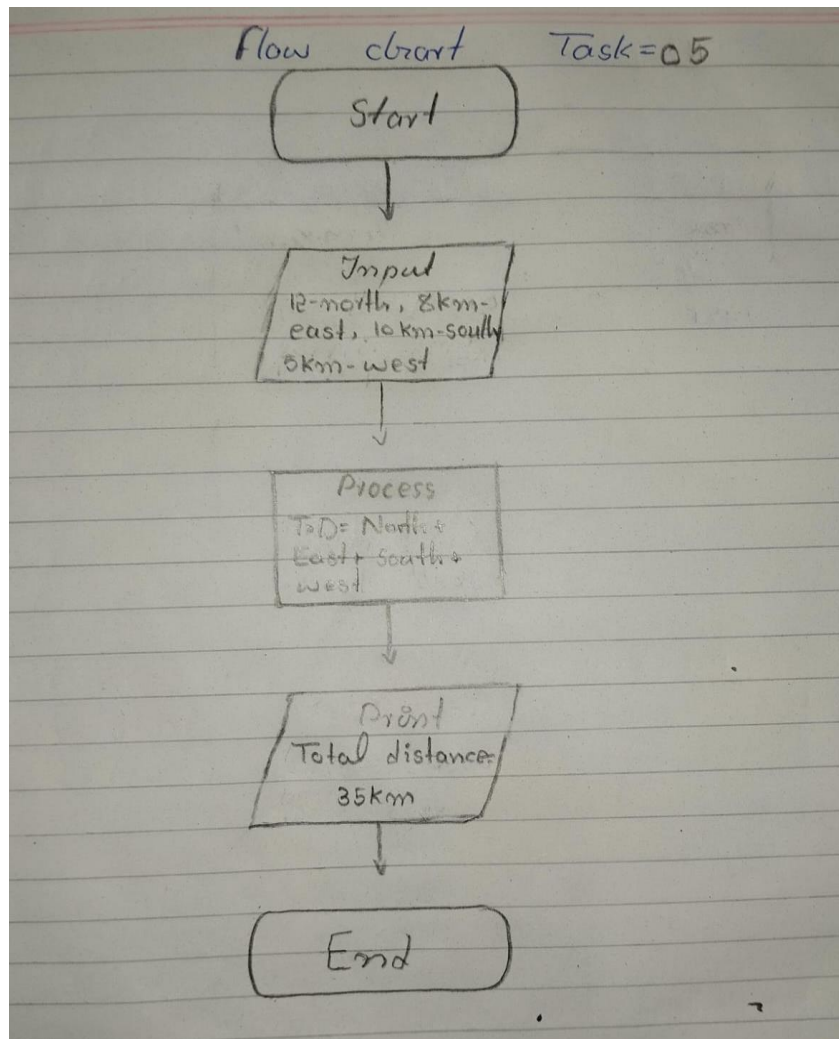
1. Create a PAC.

PAC Task = 05	
Given data	Required-Result
12 km north	• Total
8 km East	distance
10 km South	
5 km West	
Processing - required	Solution - aftermd
• Total distance =	Total distance =
north + east +	35 km
South + West	
6 km	
distance	

TASK

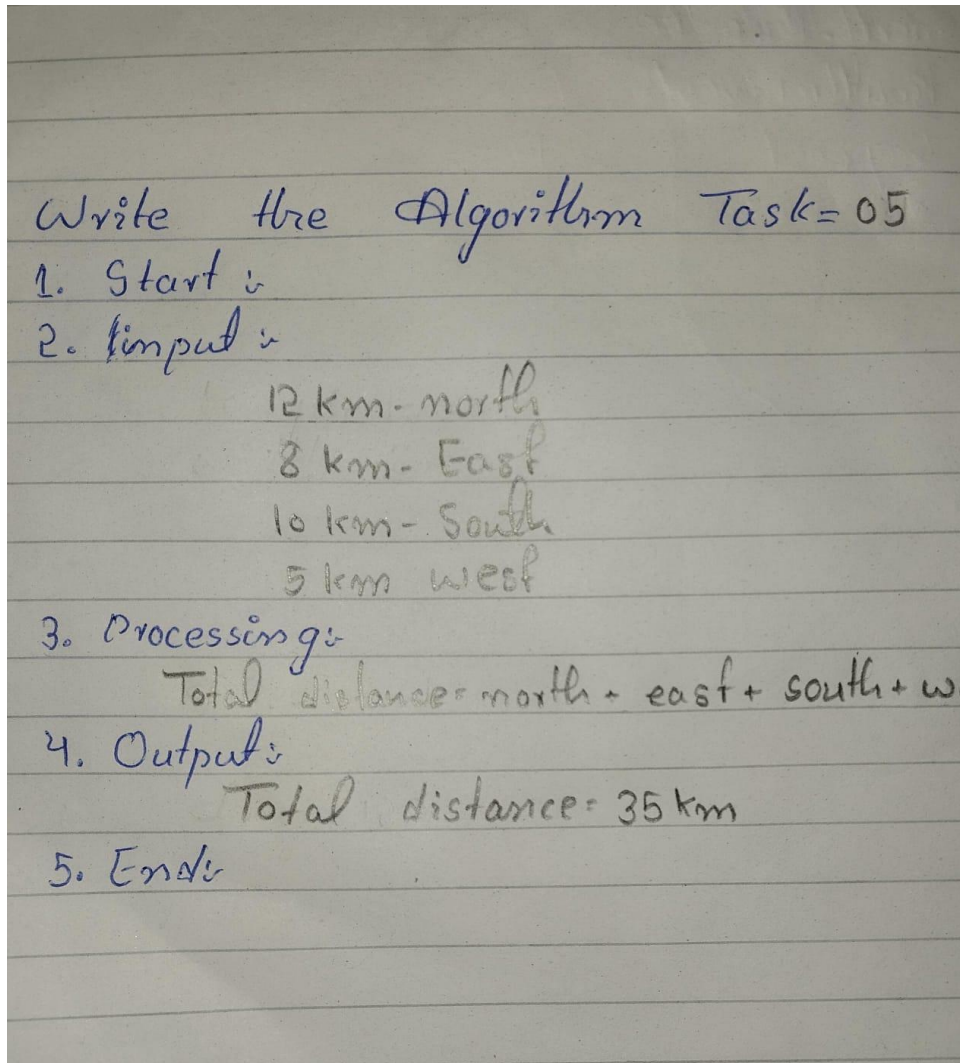
A

2. Create Flow Chart



TASK

A

3. Write an Algorithm.

TASK

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A

06 Average Marks

student received the marks in five subjects: 80, 75, 90, 85, 70 Find the Average Marks

1. Create a PAC.

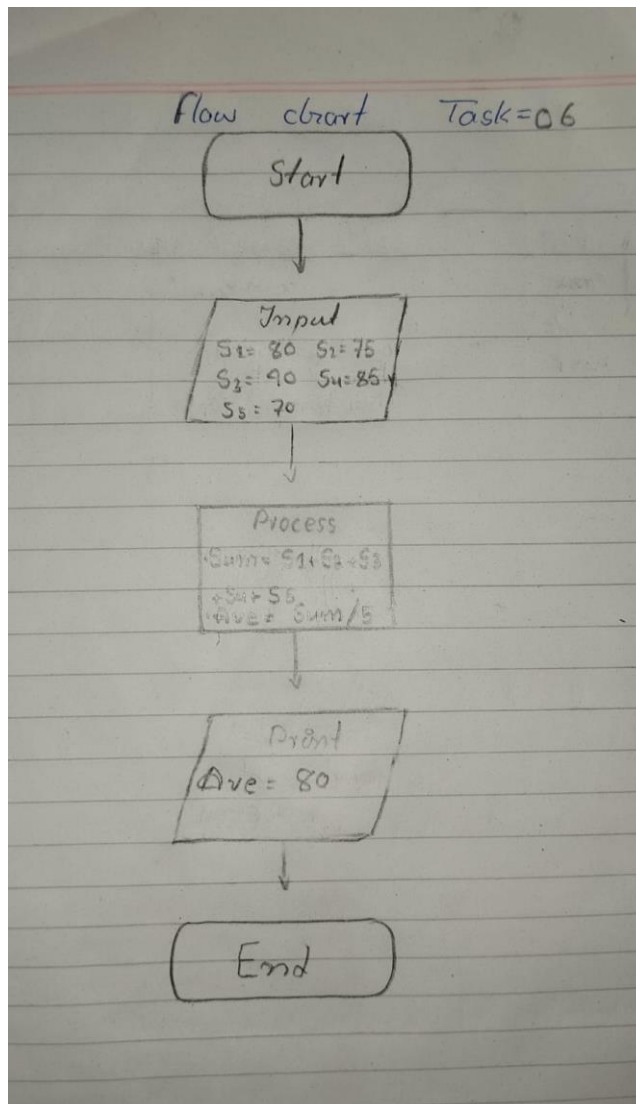
PAC Task=06

Given data	Required-Result
Sub 1 = 80	• Average
Sub 2 = 75	Marks
Sub 3 = 90	
Sub 4 = 85	
Sub 5 = 70	
Processing -required	Solution-alternate
• Sum = S1 + S2 + S3 + S4 + S5	Average = 80
• Ave = Sum / 5	

TASK

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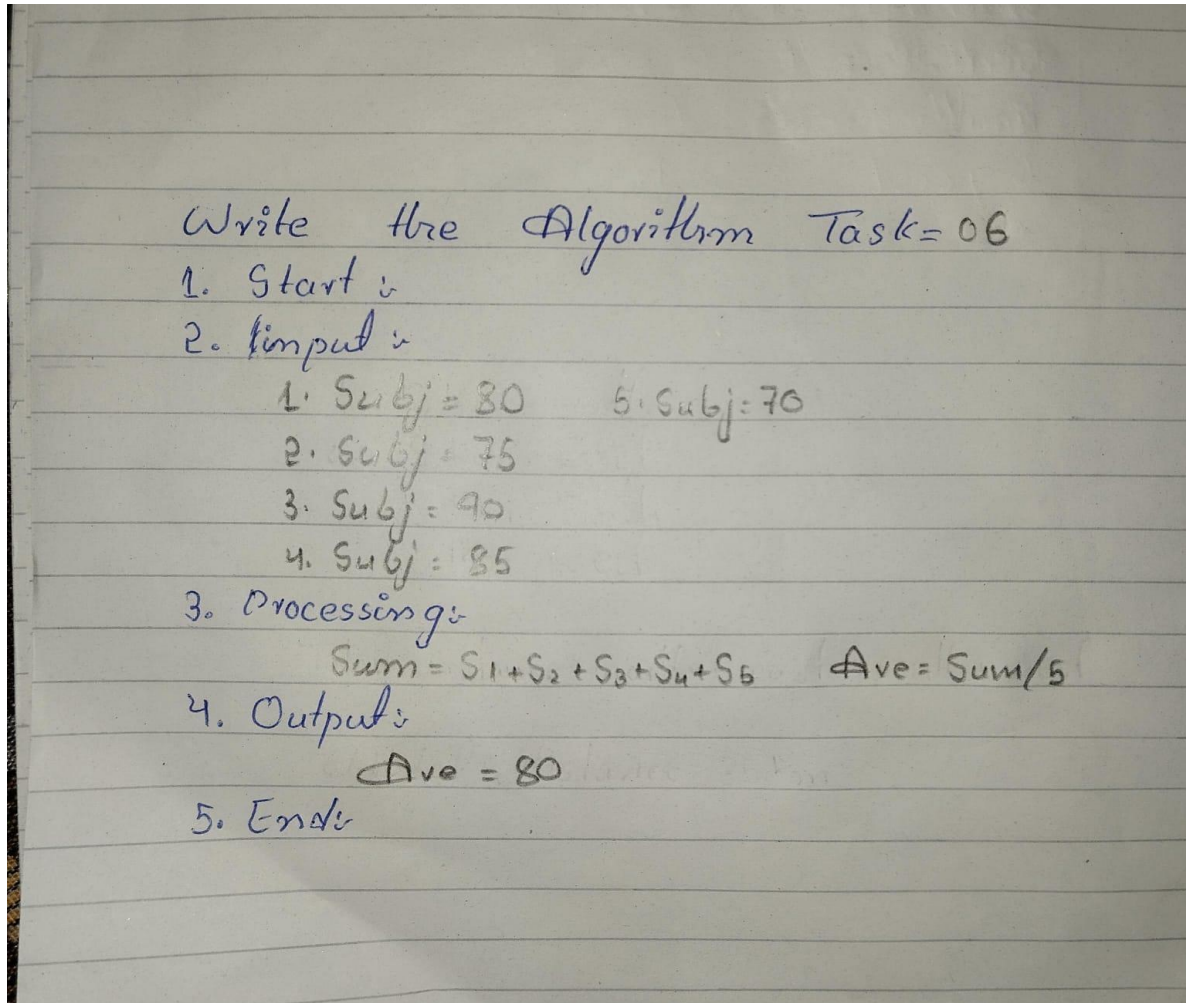
2. Create Flow Chart



TASK

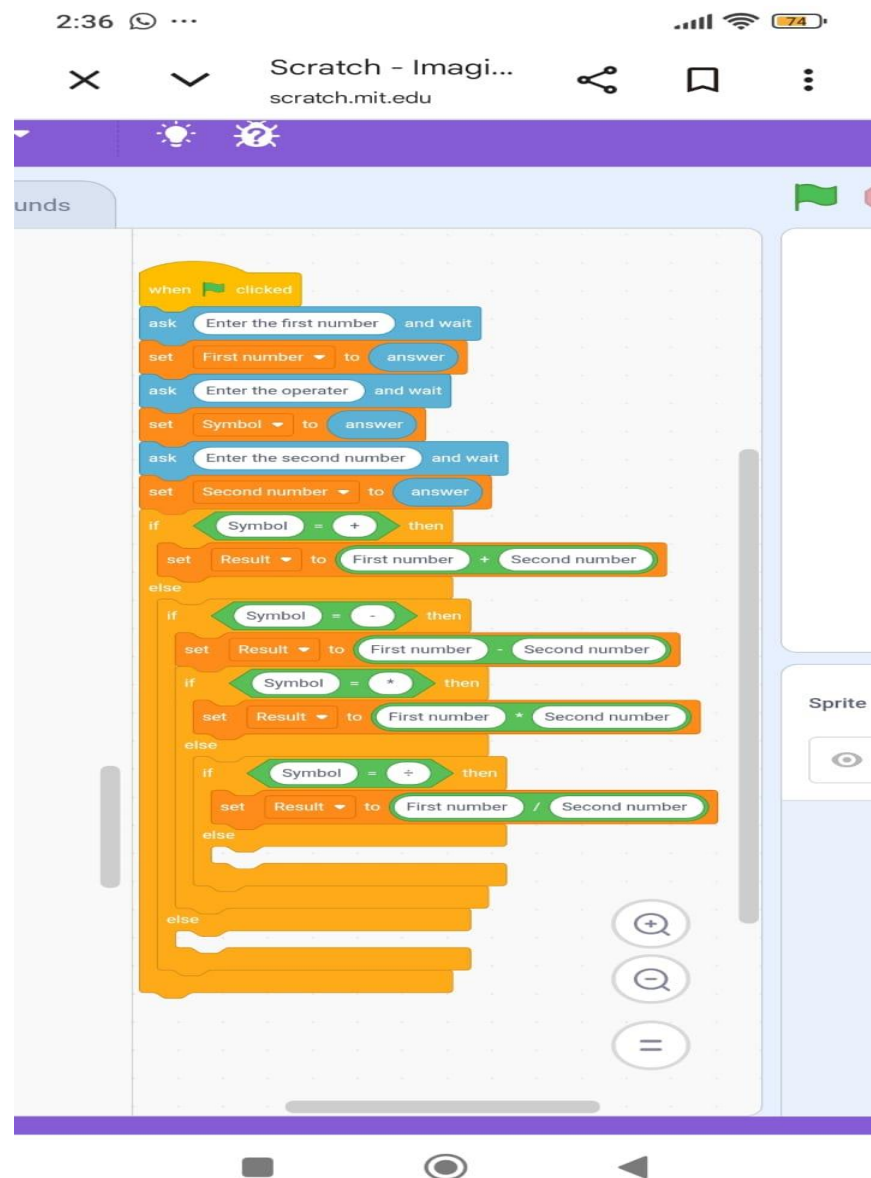
A

3. Write an Algorithm.



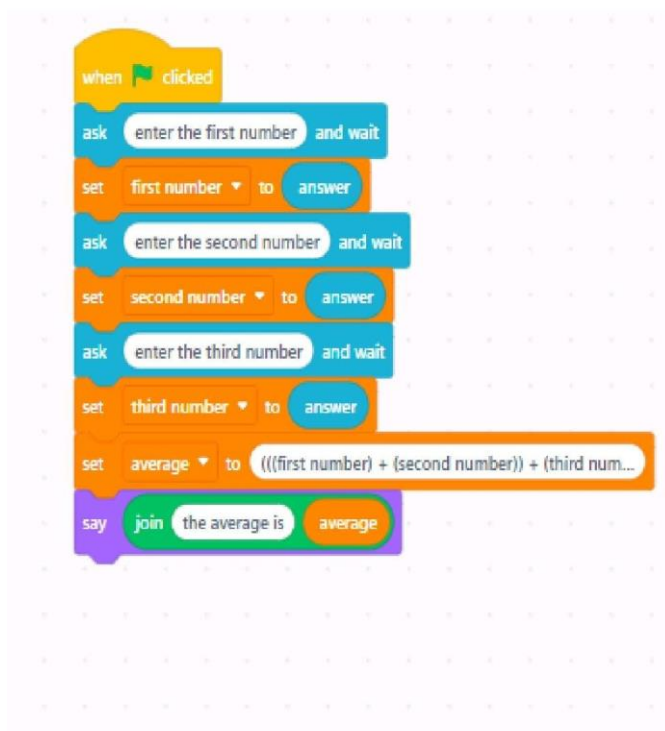
TASK # — Scratch Problem**07**

Make a simple calculator which performs basic arithmetic operations of mathematics such as addition, subtraction, division and multiplication using scratch .

SCREENSHOT

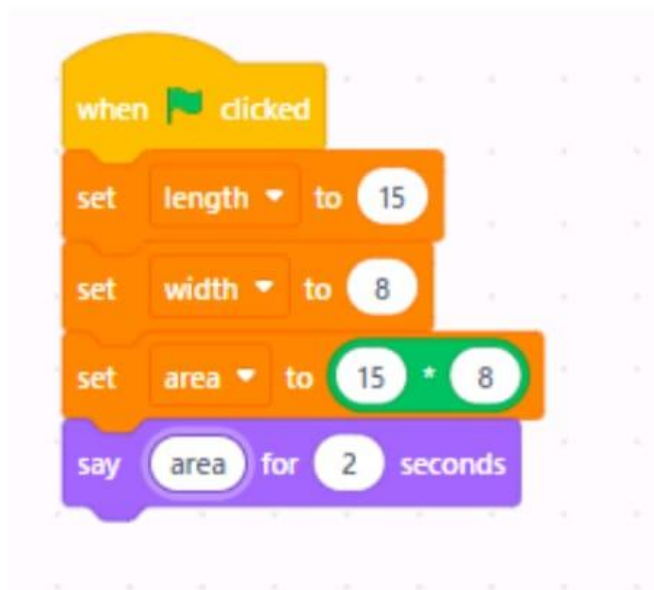
TASK # — Scratch Problem**08**

Create a Scratch program that asks the user for three numbers, calculates their average, and displays the result.

SCREENSHOT

TASK # — Scratch Problem**09**

A rectangular classroom board has a **length of 15 cm** and a **width of 8 cm**. Find Rectangle Area

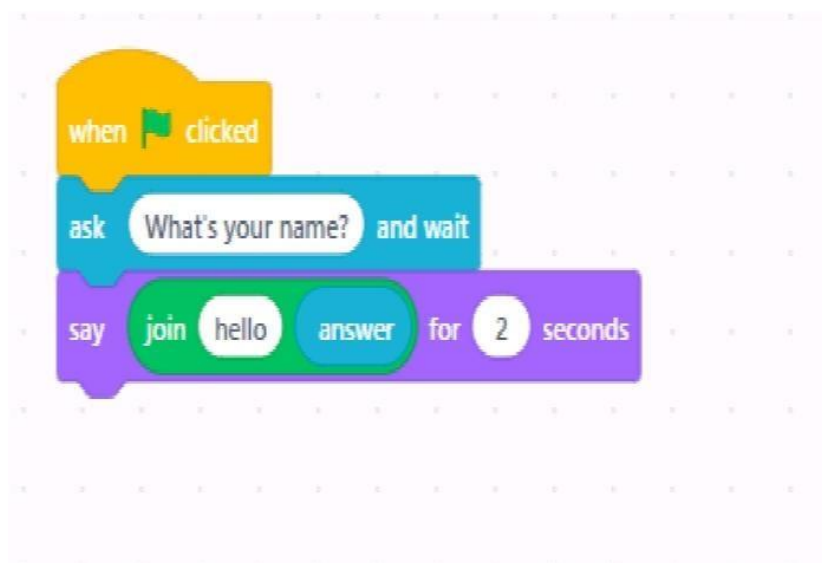
SCREENSHOT

TASK # — Scratch Problem

10

Create a Scratch program that asks the user for their name and displays a greeting message using the entered name.

Example: Hello! Welcome to Programming Fundamentals.

SCREENSHOT

TASK # — Scratch Problem

Good luck — remember to convert to PDF before submitting.